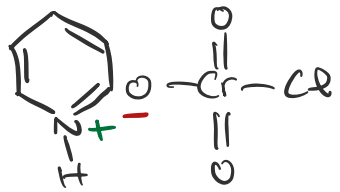
alcohol $\xrightarrow{ox}$ aldehyde/ketone $\xrightarrow{ox}$ carboxyl

PCC:  (mild oxidant)

$CH_3OH \xrightarrow{H_2Cr_2O_7} CO_2$

Mild Oxidants:

PCC

$^-OCl/CH_3COOH$ 0°C

Tempo

Swern: $DMSO/(ClCO)_2/Et_3N/CH_2Cl_2$

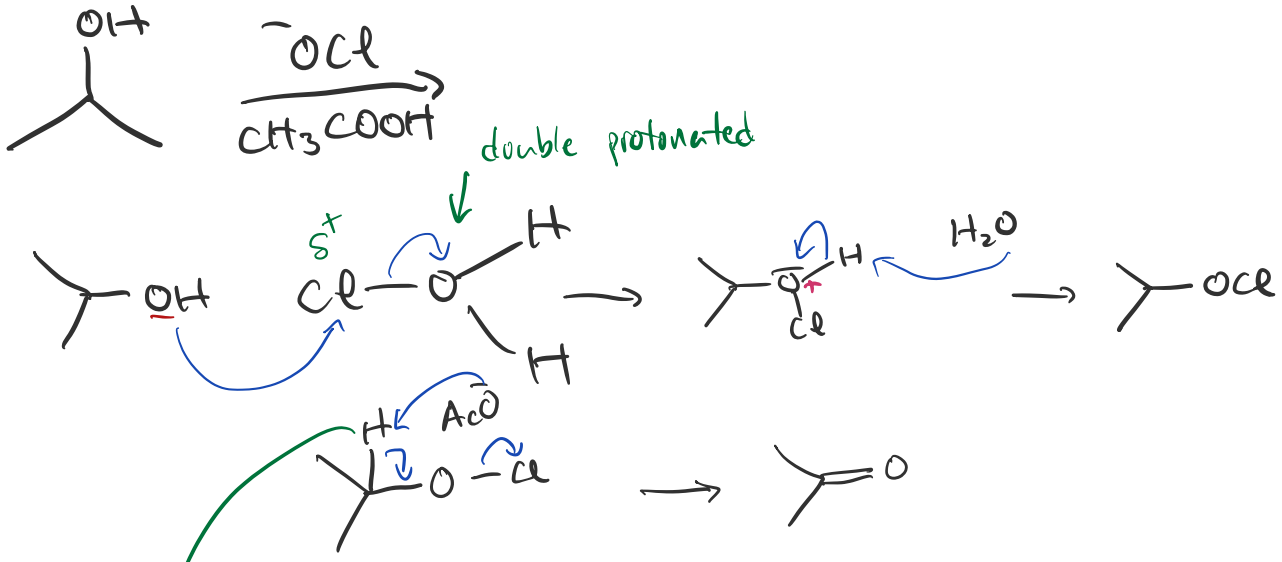
Strong:

$KMnO_4$

CrO_3/H_2SO_4

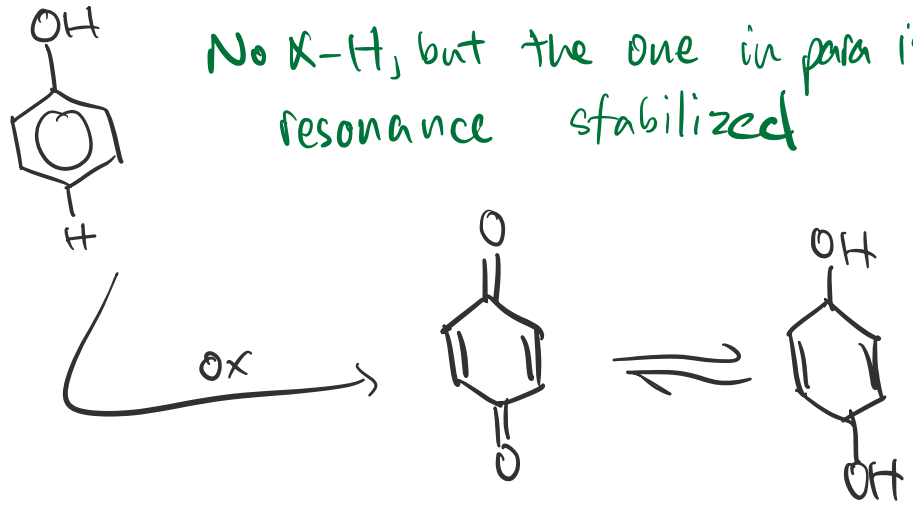
CrO_3-Pyr_2

Oxidation Mechanism



explains why α-H is needed, $3^\circ \xrightarrow{ox} X$

Phenol



No α-H, but the one in para is resonance stabilized